NVIDA GPUs have a Compute Compatibility (CC) number. This basically tells you what the GPU supports. It's a major and minor version. You can find a list of all of them online.

Cuda toolkit vs nvidia driver.

The NVIDIA driver is thought of as the "Operating system" of the GPU. The driver is a software component which has to be installed on the host systems OS. It's responsible for all GPU uses.

The NVIDIA toolkit is a set of libraries, headers and tools for writing, building and analyzing software which utlizes GPUs.

The CUDA runtime is a special case of one of the toolkit libraries. It provides an API to handle language extensions like memory allocation and copyign data between devices/hosts.

The CUDA runtime is implemented on top of the lower level API called the CUDA driver API. Which is an API exposed by the NVIDIA driver.

Most of this guide focuses on APIs exposed by the CUDA runtime. You can achieve all the same functionality by using the driver API if desired. Some features are only available using the driver API.

There is a very nice table describing memory, scopes, and lifetimes of that memory.

Memory Type	Scope	Lifetime	Location
Global	Grid	Application	Device
Constant	Grid	Application	Device
Shared	Block	Kernel	SM
Local	Thread	Kernel	Device
Register	Thread	Kernel	SM

Global memory or device memory, is the primary space for storing data that is accessible by all threads in a kernel.

Global memory is persistent. The allocation made in global memory and the data stored persists until its freed or the application ends.

Global memory is allocated with CUDA API calls such as cudaMalloc and cudaMallocManaged. And copied between host/device with cudaMemcpy.

Because global memory is accessible by all threads in a grid, you must be careful to avoid data races between threads.

Shared memory

Shared memory is accessible by all threads in a thread block. You must still be careful about data races within a thread block when using shared memory.

Shared memory is located on an SM and uses the same resources as the L1 cache, the unified data cache.

Data in shared memory persists through kernel lifetime.

Shared memory is smaller but it is located on each SM, so the bandwidth is higher and the latency is lower compared to accessing global memory.

Static allocation of shared memory.

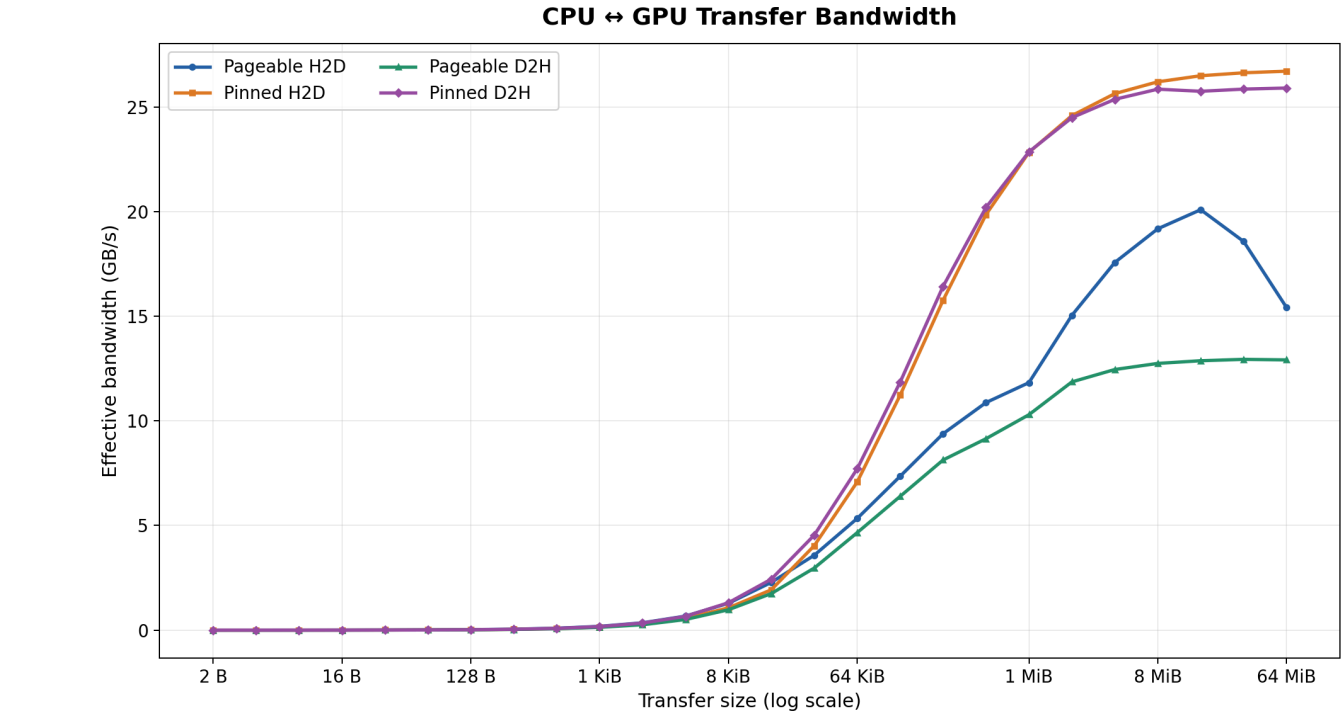
To allocate shared memory statically, you use __shared__ as a specifier on the variable inside your kernel.

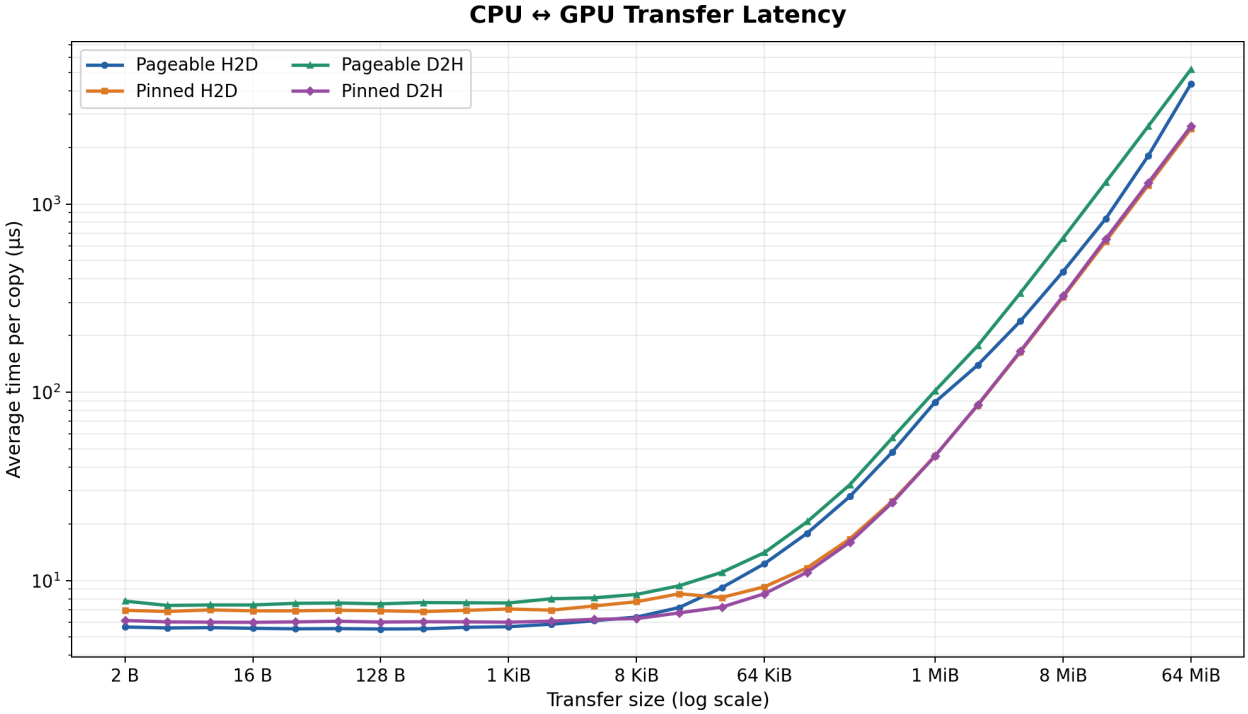
You can also specify dynamic shared memory by using the third parameter when launching a kernel. Otherwise it defaults to 0.

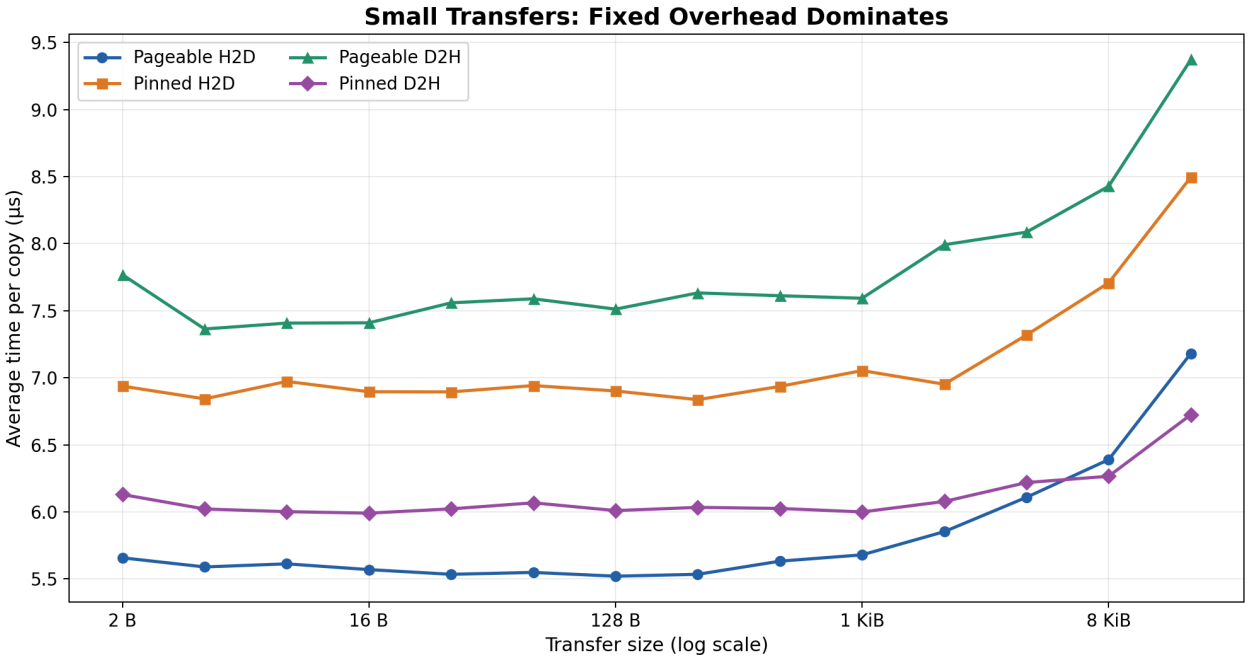
Registers

Registeres are located on the SM and have thread local scope. A register usage is managed by the compiler and registered are used for thread local storage during the execution of a kernel.

Benchmark notes


CPU ↔ GPU Transfer Bandwidth
Pageable H2D
Pinned H2D
Pageable D2H
Pinned D2H
Effective bandwidth (GB/s)
0
5
10
15
20
25
2 B
16 B
128 B
1 KiB
8 KiB
64 KiB
1 MiB
8 MiB
64 MiB
Transfer size (log scale)


CPU ↔ GPU Transfer Latency
Pageable H2D
Pinned H2D
Pageable D2H
Pinned D2H
Average time per copy (µs)
10¹
10²
10³
2 B
16 B
128 B
1 KiB
8 KiB
64 KiB
1 MiB
8 MiB
64 MiB
Transfer size (log scale)



raw data output:

Latency Bandwidth

[256 bytes, 1 chunks] Pageable Host to Device: 6.097 us, 0.042 GB/s
[512 bytes, 1 chunks] Pageable Host to Device: 6.001 us, 0.085 GB/s
[1024 bytes, 1 chunks] Pageable Host to Device: 6.015 us, 0.170 GB/s
[2048 bytes, 1 chunks] Pageable Host to Device: 6.203 us, 0.330 GB/s
[4096 bytes, 1 chunks] Pageable Host to Device: 6.527 us, 0.628 GB/s
[8192 bytes, 1 chunks] Pageable Host to Device: 6.792 us, 1.206 GB/s
[16384 bytes, 1 chunks] Pageable Host to Device: 7.820 us, 2.095 GB/s
[32768 bytes, 1 chunks] Pageable Host to Device: 9.764 us, 3.356 GB/s
[65536 bytes, 1 chunks] Pageable Host to Device: 13.221 us, 4.957 GB/s
[131072 bytes, 1 chunks] Pageable Host to Device: 18.280 us, 7.170 GB/s
[262144 bytes, 1 chunks] Pageable Host to Device: 28.356 us, 9.245 GB/s
[524288 bytes, 1 chunks] Pageable Host to Device: 48.860 us, 10.730 GB/s
[1048576 bytes, 1 chunks] Pageable Host to Device: 88.973 us, 11.785 GB/s
[2097152 bytes, 1 chunks] Pageable Host to Device: 140.113 us, 14.968 GB/s
[4194304 bytes, 1 chunks] Pageable Host to Device: 242.940 us, 17.265 GB/s
[8388608 bytes, 1 chunks] Pageable Host to Device: 449.893 us, 18.646 GB/s
[16777216 bytes, 1 chunks] Pageable Host to Device: 864.057 us, 19.417 GB/s
[33554432 bytes, 1 chunks] Pageable Host to Device: 1749.397 us, 19.181 GB/s
[67108864 bytes, 1 chunks] Pageable Host to Device: 4287.170 us, 15.653 GB/s
[256 bytes, 1 chunks] Pinned Host to Device: 6.854 us, 0.037 GB/s

[512 bytes, 1 chunks] Pinned Host to Device: 7.073 us, 0.072 GB/s
[1024 bytes, 1 chunks] Pinned Host to Device: 6.854 us, 0.149 GB/s
[2048 bytes, 1 chunks] Pinned Host to Device: 7.247 us, 0.283 GB/s
[4096 bytes, 1 chunks] Pinned Host to Device: 7.685 us, 0.533 GB/s
[8192 bytes, 1 chunks] Pinned Host to Device: 7.556 us, 1.084 GB/s
[16384 bytes, 1 chunks] Pinned Host to Device: 8.371 us, 1.957 GB/s
[32768 bytes, 1 chunks] Pinned Host to Device: 8.006 us, 4.093 GB/s
[65536 bytes, 1 chunks] Pinned Host to Device: 9.127 us, 7.180 GB/s
[131072 bytes, 1 chunks] Pinned Host to Device: 11.599 us, 11.301 GB/s
[262144 bytes, 1 chunks] Pinned Host to Device: 16.489 us, 15.898 GB/s
[524288 bytes, 1 chunks] Pinned Host to Device: 26.308 us, 19.929 GB/s
[1048576 bytes, 1 chunks] Pinned Host to Device: 45.835 us, 22.877 GB/s
[2097152 bytes, 1 chunks] Pinned Host to Device: 85.171 us, 24.623 GB/s
[4194304 bytes, 1 chunks] Pinned Host to Device: 163.447 us, 25.662 GB/s
[8388608 bytes, 1 chunks] Pinned Host to Device: 319.971 us, 26.217 GB/s
[16777216 bytes, 1 chunks] Pinned Host to Device: 633.040 us, 26.503 GB/s
[33554432 bytes, 1 chunks] Pinned Host to Device: 1259.100 us, 26.650 GB/s
[67108864 bytes, 1 chunks] Pinned Host to Device: 2511.666 us, 26.719 GB/s
[256 bytes, 1 chunks] Pageable Device to Host: 7.950 us, 0.032 GB/s
[512 bytes, 1 chunks] Pageable Device to Host: 7.905 us, 0.065 GB/s
[1024 bytes, 1 chunks] Pageable Device to Host: 7.704 us, 0.133 GB/s
[2048 bytes, 1 chunks] Pageable Device to Host: 7.866 us, 0.260 GB/s
[4096 bytes, 1 chunks] Pageable Device to Host: 8.128 us, 0.504 GB/s
[8192 bytes, 1 chunks] Pageable Device to Host: 8.373 us, 0.978 GB/s
[16384 bytes, 1 chunks] Pageable Device to Host: 9.163 us, 1.788 GB/s
[32768 bytes, 1 chunks] Pageable Device to Host: 10.917 us, 3.002 GB/s
[65536 bytes, 1 chunks] Pageable Device to Host: 13.972 us, 4.691 GB/s
[131072 bytes, 1 chunks] Pageable Device to Host: 20.318 us, 6.451 GB/s
[262144 bytes, 1 chunks] Pageable Device to Host: 32.366 us, 8.099 GB/s
[524288 bytes, 1 chunks] Pageable Device to Host: 59.950 us, 8.745 GB/s
[1048576 bytes, 1 chunks] Pageable Device to Host: 108.361 us, 9.677 GB/s
[2097152 bytes, 1 chunks] Pageable Device to Host: 182.092 us, 11.517 GB/s
[4194304 bytes, 1 chunks] Pageable Device to Host: 339.883 us, 12.340 GB/s
[8388608 bytes, 1 chunks] Pageable Device to Host: 661.793 us, 12.676 GB/s
[16777216 bytes, 1 chunks] Pageable Device to Host: 1308.459 us, 12.822 GB/s
[33554432 bytes, 1 chunks] Pageable Device to Host: 2591.466 us, 12.948 GB/s
[67108864 bytes, 1 chunks] Pageable Device to Host: 5165.812 us, 12.991 GB/s
[256 bytes, 1 chunks] Pinned Device to Host: 6.139 us, 0.042 GB/s
[512 bytes, 1 chunks] Pinned Device to Host: 6.077 us, 0.084 GB/s
[1024 bytes, 1 chunks] Pinned Device to Host: 6.068 us, 0.169 GB/s
[2048 bytes, 1 chunks] Pinned Device to Host: 6.168 us, 0.332 GB/s
[4096 bytes, 1 chunks] Pinned Device to Host: 6.240 us, 0.656 GB/s
[8192 bytes, 1 chunks] Pinned Device to Host: 6.371 us, 1.286 GB/s
[16384 bytes, 1 chunks] Pinned Device to Host: 6.615 us, 2.477 GB/s

[32768 bytes, 1 chunks] Pinned Device to Host: 7.277 us, 4.503 GB/s
[65536 bytes, 1 chunks] Pinned Device to Host: 8.436 us, 7.768 GB/s
[131072 bytes, 1 chunks] Pinned Device to Host: 11.030 us, 11.883 GB/s
[262144 bytes, 1 chunks] Pinned Device to Host: 16.072 us, 16.311 GB/s
[524288 bytes, 1 chunks] Pinned Device to Host: 25.950 us, 20.204 GB/s
[1048576 bytes, 1 chunks] Pinned Device to Host: 45.835 us, 22.877 GB/s
[2097152 bytes, 1 chunks] Pinned Device to Host: 85.695 us, 24.472 GB/s
[4194304 bytes, 1 chunks] Pinned Device to Host: 165.367 us, 25.364 GB/s
[8388608 bytes, 1 chunks] Pinned Device to Host: 324.458 us, 25.854 GB/s
[16777216 bytes, 1 chunks] Pinned Device to Host: 643.259 us, 26.082 GB/s
[33554432 bytes, 1 chunks] Pinned Device to Host: 1280.502 us, 26.204 GB/s
[67108864 bytes, 1 chunks] Pinned Device to Host: 2554.408 us, 26.272 GB/s
[256 bytes, 32 chunks] Pageable Host to Device: 190.267 us, 0.001 GB/s
[512 bytes, 32 chunks] Pageable Host to Device: 189.414 us, 0.003 GB/s
[1024 bytes, 32 chunks] Pageable Host to Device: 189.600 us, 0.005 GB/s
[2048 bytes, 32 chunks] Pageable Host to Device: 190.739 us, 0.011 GB/s
[4096 bytes, 32 chunks] Pageable Host to Device: 189.831 us, 0.022 GB/s
[8192 bytes, 32 chunks] Pageable Host to Device: 191.107 us, 0.043 GB/s
[16384 bytes, 32 chunks] Pageable Host to Device: 192.935 us, 0.085 GB/s
[32768 bytes, 32 chunks] Pageable Host to Device: 195.320 us, 0.168 GB/s
[65536 bytes, 32 chunks] Pageable Host to Device: 199.503 us, 0.328 GB/s
[131072 bytes, 32 chunks] Pageable Host to Device: 207.756 us, 0.631 GB/s
[262144 bytes, 32 chunks] Pageable Host to Device: 220.260 us, 1.190 GB/s
[524288 bytes, 32 chunks] Pageable Host to Device: 248.140 us, 2.113 GB/s
[1048576 bytes, 32 chunks] Pageable Host to Device: 300.630 us, 3.488 GB/s
[2097152 bytes, 32 chunks] Pageable Host to Device: 379.954 us, 5.519 GB/s
[4194304 bytes, 32 chunks] Pageable Host to Device: 557.647 us, 7.521 GB/s
[8388608 bytes, 32 chunks] Pageable Host to Device: 880.806 us, 9.524 GB/s
[16777216 bytes, 32 chunks] Pageable Host to Device: 1538.660 us, 10.904 GB/s
[33554432 bytes, 32 chunks] Pageable Host to Device: 2851.640 us, 11.767 GB/s
[67108864 bytes, 32 chunks] Pageable Host to Device: 5088.537 us, 13.188 GB/s
[256 bytes, 32 chunks] Pinned Host to Device: 221.743 us, 0.001 GB/s
[512 bytes, 32 chunks] Pinned Host to Device: 220.585 us, 0.002 GB/s
[1024 bytes, 32 chunks] Pinned Host to Device: 221.461 us, 0.005 GB/s
[2048 bytes, 32 chunks] Pinned Host to Device: 220.960 us, 0.009 GB/s
[4096 bytes, 32 chunks] Pinned Host to Device: 221.147 us, 0.019 GB/s
[8192 bytes, 32 chunks] Pinned Host to Device: 221.009 us, 0.037 GB/s
[16384 bytes, 32 chunks] Pinned Host to Device: 221.859 us, 0.074 GB/s
[32768 bytes, 32 chunks] Pinned Host to Device: 225.711 us, 0.145 GB/s
[65536 bytes, 32 chunks] Pinned Host to Device: 250.231 us, 0.262 GB/s
[131072 bytes, 32 chunks] Pinned Host to Device: 256.912 us, 0.510 GB/s
[262144 bytes, 32 chunks] Pinned Host to Device: 271.658 us, 0.965 GB/s
[524288 bytes, 32 chunks] Pinned Host to Device: 300.371 us, 1.745 GB/s
[1048576 bytes, 32 chunks] Pinned Host to Device: 280.824 us, 3.734 GB/s

[2097152 bytes, 32 chunks] Pinned Host to Device: 308.848 us, 6.790 GB/s
[4194304 bytes, 32 chunks] Pinned Host to Device: 373.606 us, 11.227 GB/s
[8388608 bytes, 32 chunks] Pinned Host to Device: 528.997 us, 15.858 GB/s
[16777216 bytes, 32 chunks] Pinned Host to Device: 842.095 us, 19.923 GB/s
[33554432 bytes, 32 chunks] Pinned Host to Device: 1468.389 us, 22.851 GB/s
[67108864 bytes, 32 chunks] Pinned Host to Device: 2722.486 us, 24.650 GB/s
[256 bytes, 32 chunks] Pageable Device to Host: 239.635 us, 0.001 GB/s
[512 bytes, 32 chunks] Pageable Device to Host: 244.995 us, 0.002 GB/s
[1024 bytes, 32 chunks] Pageable Device to Host: 245.104 us, 0.004 GB/s
[2048 bytes, 32 chunks] Pageable Device to Host: 242.519 us, 0.008 GB/s
[4096 bytes, 32 chunks] Pageable Device to Host: 246.737 us, 0.017 GB/s
[8192 bytes, 32 chunks] Pageable Device to Host: 248.042 us, 0.033 GB/s
[16384 bytes, 32 chunks] Pageable Device to Host: 248.505 us, 0.066 GB/s
[32768 bytes, 32 chunks] Pageable Device to Host: 256.224 us, 0.128 GB/s
[65536 bytes, 32 chunks] Pageable Device to Host: 273.206 us, 0.240 GB/s
[131072 bytes, 32 chunks] Pageable Device to Host: 283.328 us, 0.463 GB/s
[262144 bytes, 32 chunks] Pageable Device to Host: 294.661 us, 0.890 GB/s
[524288 bytes, 32 chunks] Pageable Device to Host: 323.994 us, 1.618 GB/s
[1048576 bytes, 32 chunks] Pageable Device to Host: 380.396 us, 2.757 GB/s
[2097152 bytes, 32 chunks] Pageable Device to Host: 529.664 us, 3.959 GB/s
[4194304 bytes, 32 chunks] Pageable Device to Host: 847.121 us, 4.951 GB/s
[8388608 bytes, 32 chunks] Pageable Device to Host: 1459.486 us, 5.748 GB/s
[16777216 bytes, 32 chunks] Pageable Device to Host: 3193.874 us, 5.253 GB/s
[33554432 bytes, 32 chunks] Pageable Device to Host: 6038.339 us, 5.557 GB/s
[67108864 bytes, 32 chunks] Pageable Device to Host: 10849.145 us, 6.186 GB/s
[256 bytes, 32 chunks] Pinned Device to Host: 215.575 us, 0.001 GB/s
[512 bytes, 32 chunks] Pinned Device to Host: 217.089 us, 0.002 GB/s
[1024 bytes, 32 chunks] Pinned Device to Host: 216.358 us, 0.005 GB/s
[2048 bytes, 32 chunks] Pinned Device to Host: 217.241 us, 0.009 GB/s
[4096 bytes, 32 chunks] Pinned Device to Host: 216.421 us, 0.019 GB/s
[8192 bytes, 32 chunks] Pinned Device to Host: 217.057 us, 0.038 GB/s
[16384 bytes, 32 chunks] Pinned Device to Host: 216.850 us, 0.076 GB/s
[32768 bytes, 32 chunks] Pinned Device to Host: 217.900 us, 0.150 GB/s
[65536 bytes, 32 chunks] Pinned Device to Host: 219.471 us, 0.299 GB/s
[131072 bytes, 32 chunks] Pinned Device to Host: 223.866 us, 0.585 GB/s
[262144 bytes, 32 chunks] Pinned Device to Host: 228.539 us, 1.147 GB/s
[524288 bytes, 32 chunks] Pinned Device to Host: 244.435 us, 2.145 GB/s
[1048576 bytes, 32 chunks] Pinned Device to Host: 265.025 us, 3.957 GB/s
[2097152 bytes, 32 chunks] Pinned Device to Host: 296.630 us, 7.070 GB/s
[4194304 bytes, 32 chunks] Pinned Device to Host: 356.271 us, 11.773 GB/s
[8388608 bytes, 32 chunks] Pinned Device to Host: 514.120 us, 16.316 GB/s
[16777216 bytes, 32 chunks] Pinned Device to Host: 833.277 us, 20.134 GB/s
[33554432 bytes, 32 chunks] Pinned Device to Host: 1481.360 us, 22.651 GB/s
[67108864 bytes, 32 chunks] Pinned Device to Host: 2754.909 us, 24.360 GB/s